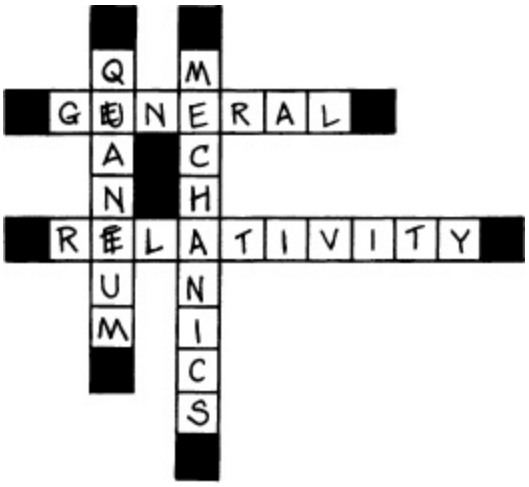


Progress was also slow on trying to understand the fundamental building blocks of atomic nuclei—the neutron, the proton, the electron, and the plethora of other *elementary particles* that were being created in particle accelerators.

Wheeler had a dream of leaping over these impasses and catching a simultaneous glimpse of the nature of quantum gravity and the nature of elementary particles. Such a glimpse, he thought, might come from seeking out those places in theoretical physics where paradoxes abound. From resolving a paradox comes deep understanding. The deeper the paradox, the more likely that the understanding would probe beyond twentieth-century frontiers.

It was in this spirit that, soon after emerging from the superbomb effort, Wheeler, with Harrison and Wakano, filled in the missing gaps in our knowledge of cold, dead stars (Chapter 5); and it was in this spirit that Wheeler contemplated the resulting "fate of great masses." Here was a deep paradox of just the sort Wheeler was seeking: No cold, dead star can be more massive than about 2 Suns; and yet the heavens seem to abound in hot stars far more massive than that—stars which some day must cool and die. Oppenheimer, in his straightforward way, had asked the well-established laws of physics what happens to such stars, and had got (with Snyder) an answer that seemed outrageous to Wheeler. This reinforced Wheeler's conviction that here, in the fates of great masses, he might catch a glimpse of physics beyond twentieth-century frontiers. Wheeler was right, as we shall see in Chapters 12 and 13.

Wheeler had fire in his belly—a deep, unremitting need to know the fate of great masses and learn whether their fate might unlock the mysteries of